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Fri 17 Jul 2026 16:34:18 - sadmlib_std_demo.py v03.28.00 - Library v4.69.4

Short description of program

bilbo.maison.ca - User: root - Umask: 0002 - Arch: x86_64

Fedora Linux v43 - Kernel 7.1.3-101.fc43.x86_64 - SADMIN(C): /opt/sadmin

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01) SADMIN Python Library Functions Available To Developers

[001]	sa.get_release()	SADMIN Release Number (XX.XX)	2.1.2
[002]	sa.get_ostype()	OS Type (In Uppercase,LINUX,AIX,MACOS)	LINUX
[003]	sa.get_osversion()	Return O/S Version (Ex: 7.2, 6.5)	43
[004]	sa.get_osmajorversion()	Return O/S Major Version (Ex 7, 6)	43
[005]	sa.get_osminorversion()	Return O/S Minor Version (Ex 2, 3)	
[006]	sa.get_osname()	O/S Name (REDHAT,ROCKY,DEBIAN,...)	FEDORA
[007]	sa.get_oscodename()	O/S Project Code Name	N/A
[008]	sa.get_username()	Return Current User Name	root
[009]	sa.get_kernel_version()	Return O/S Running Kernel Version	7.1.3-101.fc43.x86_64
[010]	sa.get_kernel_bitmode()	Return O/S Kernel Bit Mode (32 or 64)	64
[011]	sa.get_hostname()	Return Host Name	bilbo
[012]	sa.get_host_ip()	Return Host Main IP Address	192.168.1.23
[013]	sa.get_domainname()	Return Host Domain Name	maison.ca
[014]	sa.get_fqdn()	Return Fully Qualified Domain Name	bilbo.maison.ca
[015]	sa.get_serial()	Return System serial Number	L2HF78K0045
[016]	sa.get_server_cpu_speed()	Return Maximum CPU Speed	3000 MHz
[017]	sa.get_nb_cpu	Return number of Physical CPUs	2
[018]	sa.get_nb_logical_cpu()	Return number of Logical CPUs	4
[019]	sa.get_nb_core_per_socket()	Return number of core per socket	2
[020]	sa.get_nb_thread_per_core()	Return number of threads per core	2
[021]	sa.get_nb_socket()	Return number of socket	1
[022]	sa.get_packagetype()	Return package type (rpm,deb,lpp,dmg)	rpm
[023]	sa.get_arch()	Get system architecture	x86_64
[024]	sa.lock_system(hostname)	Lock the specified hostname (Not FQDN)	True
[025]	sa.lock_status(hostname)	Check if host specified is lock	True
[026]	sa.unlock_system(hostname)	Unlock host specified, monitoring on	True
[027]	sa.write_log('message'[,lf=true])	Write message to Screen,Log or Both	message
[028]	sa.write_err('message'[,lf=true])	Write message to Error and Std Log	message
[029]	sa.sendmail(addr,sub,body,attach)	Send email (body=textFile) & opt.attach	0=Success 1=Error
[030]	sa.silentremove('file')	Silent file delete (no msg, no err)	0
[031]	sa.trimfile(filename,nlines=500)	Keep last 500 lines of filename.	0=Success 1=Error
[032]	sa.sleep(60,15)	Sleep 60 sec & update every 15 sec.	60...45...30...15...0
[033]	sa.get_epoch_time()	Return Current Epoch Time	1784320459
[034]	sa.epoch_to_date(1784320459)	Convert epoch time to date	2026.07.17 16:34:19
[035]	sa.date_to_epoch(wdate)	Convert Date to epoch time	1784320459
	DATE1 (End): 2026.06.30 10:00:44	DATE2 (Start): 2026.06.30 09:50:03	
[036]	sa.elapse_time (DATE1,DATE2)	Elapse Time between two timestamps	00:10:41
[037]	sa.show_version(sa.ver)	Show Program & Library version & Desc.	scriptName -v
[038]	sa.touch_file('filename')	Update timestamp or create file	0
[039]	sa.db_connect('sadmin')	Open connection to database	0=Success 1=Error
[040]	sa.db_close(conn,cur)	Close connection to database	0=Success 1=Error

02) SADMIN Section of your program that is shared with SADMIN Python Library.

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[001] pid                Current Process ID                368820
[002] hostname           Current Host Name                bilbo
[003] os_type            OS Type (In Uppercase,LINUX,AIX,MACOS)  LINUX
[004] username           Current user name                root
[005] debug              Debug Level 0-9 (Incr. Verbose)        0
[006] exit_code          Program Exit Return Code         0
[007] cmd_ssh_full       SSH cmd to client                /usr/bin/ssh -qnp 32

[008] sa.pn              Program name with extension      sadmlib_std_demo.py
[009] sa.inst            Program name without extension    sadmlib_std_demo
[010] sa.ver             Program version number           03.28.00
[011] sa.desc            Program description              Short description of program
[012] sa.root_only       Program can only be run by 'root' False
[013] sa.server_only     Program can only run on SADMIN server False
[014] sa.sadm_group_only Run only if part of SADMIN Grp or root False
[015] sa.multiple_exec   Can run more than 1 copy simultaneously False
[016] sa.quiet           Quiet False=ErrNo+Mess True=OnlyErrNo False
[017] sa.log_type        Send Output to [S]creen [L]og [B]oth B
[018] sa.log_append      True=Append to log, False=New log False
[019] sa.log_header      Generate header in log          True
[020] sa.log_footer      Generate footer in log          True
[021] sa.use_rch          Save execution data to '.rch' file True
[022] sa.errno           Error No. returned by function called 0
[023] sa.errmsg          Error Mess. returned by function called
[024] sa.sadm_pid_timeout PID file default TimeToLive (Sec) 14400 sec
[025] sa.sadm_lock_timeout Maximun nb. sec. a host can be lock 7200 sec
[026] sa.db_used         Need Access to SADMIN Database   False
[027] sa.db_name         Database Name (if not 'sadmin')   sadmin
[028] sa.db_conn         Database connector               None
[029] sa.db_cur          Database cursor                   None

## Variables that can override default value taken from $SADMIN/cfg/sadmin.cfg.
[030] sa.sadm_alert_type 0=NoMail 1=OnError 3=OnSuccess 4=All 1
[031] sa.sadm_alert_group Error Group (Default Group) default
[032] sa.sadm_warning_group Warning Group Name default
[033] sa.sadm_info_group  Info Group Name default
[034] sa.sadm_alert_repeat 0=AlertOnce or Sec. before alert repeat 0 sec
[035] sa.sadm_mail_addr   SADMIN Administrator Email(s) brucetalbot95@gmail.com
[036] sa.sadm_max_logline Trim log to this maximum of lines 500 lines
[037] sa.sadm_max_rchline Trim rch file to this max. of lines 30 lines
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03) Python Library - Overview of the 'start()' and 'stop()' functions.

Function 'start()'

The 'start()' function basically initialize the SADMIN environment.
When you call 'start()', it will return only if everything went OK,
Otherwise, it will advise the user of the error and exit(1).

- If 'db_used' is True, it will create a connection to the Database and return a connection (db_conn) and cursor (db_cur) objects.
Example: '(db_conn,db_cur) = sa.start()'
- If 'sa.db_used' is False, then it will return None for both 'sa_conn' and 'sadb_cur'.
Example: 'sa.start()'

Here a summary of the different things the 'start()' function does :

- 1) Make sure \$SADMIN directories and sub-directories exist and have proper permissions.
- 2) If 'sa.log_append' is True, it open standard/error log in append mode, else in write mode.
- 3) If 'sa.log_header' is True, write the log header.
- 4) If 'sa.root_only' is True and the current user is not root, show error message and exit(1).
- 5) If 'sa.server_only' is True and you are not on the SADMIN server, show error message and exit(1).
- 6) If 'sa.sadm_group_only' is 'True', then current user must be part of the SADMIN group (unless root).
- 7) If 'sa.db_used' is True, open connection to database and return connection and cursor objects.
If not on SADMIN system, show error message and exit(1).
- 8) If lock_status() is True and running on the SADMIN server, issue message and exit(1).
If the creation of the lock was done more than 'lock_timeout' seconds, the lock file is remove.
- 9) If PID file 'pid_file' exist and 'sa.multiple_exec' is True, continue normal execution.
If PID file exist and execution time (sec) is less than the 'pid_timeout', show error mess. and exit(1).
If PID file exist and execution time (sec) exceed the 'pid_timeout' a new 'pid_file' is created and execution is resume.
- 10) If 'use_rch' is True, add a line in 'rch' file with code '2' and starting time of your program.

Function 'stop()'

This function accept one parameter, either a 0 for Success or a 1 for Error.
This should be the one of the last function called at the end of your program.

What this function does:

- 1) Calculate execution Time.
- 2) If 'sa.use_rch = True', update the rch file (End Time & Elapse Time ...).
- 3) Validate & determine the alert group name (sadm_alert_group) use in rch file.
- 4) If 'sa.db_used' is True and we are on the SADMIN server, close the database connection.
- 5) If the error log file (.elog) is empty, delete it.
- 6) Delete the PID file of the script (pid_file).
- 7) If 'sa.log_footer' is True, write the log footer.
- 8) Close log and error log files.
- 9) If 'sa.max_logline' is not zero, trim the log according to user choice in 'sa.max_logline'.
- 10) If 'sa.max_rchline' is not zero, trim the 'rch' according to user choice in 'sa.max_rchline'.
- 11) Set permission and owner/group to log and rch files.
- 12) If on the SADMIN server, then rch and log are immediatly web central directory.

4) SADMIN configuration file content - List of variables available to developers.

----- General Section -----

[001]	sa.sadm_server	SADMIN server name (FQDN)	sherlock.maison.ca
[002]	sa.sadm_host_type	SADMIN [C]lient or [S]erver	C
[003]	sa.sadm_cie_name	Your Company name	The Batcave
[004]	sa.sadm_domain	Server creation default domain	maison.ca
[005]	sa.sadm_user	SADMIN User Name	sadmin
[006]	sa.sadm_group	SADMIN Group Name	sadmin
[007]	sa.sadm_pwd_random	Daily change password of '\$SADM_USER'	Y
[008]	sa.sadm_pid_timeout	PID file default TimeToLive (Sec)	14400 sec.
[009]	sa.sadm_lock_timeout	Maximun nb. sec. a host can be lock	7200 sec.

----- Database Section -----

[010]	sa.sadm_dbname	SADMIN Database Name	sadmin
[011]	sa.sadm_dbhost	SADMIN Database Host	localhost
[012]	sa.sadm_dbport	SADMIN Database Host TCP Port	3306
[013]	sa.sadm_ro_dbuser	SADMIN Database Read Only User	squery
[014]	sa.sadm_ro_dbpwd	SADMIN Database Read Only User Pwd	*Hidden*
[015]	sa.sadm_rw_dbuser	SADMIN Database Read/Write User	sadmin
[016]	sa.sadm_rw_dbpwd	SADMIN Database Read/Write User Pwd	*Hidden*
[017]	sa.dbpass_file	SADMIN Database User Password File	*Hidden*

----- Web Interface Section -----

[018]	sa.sadm_www_user	User that Run the Web Server	apache
[019]	sa.sadm_www_group	Group that Run the Web Server	apache
[020]	sa.sadm_monitor_update_interval	Monitor web page refresh rate (Sec)	60 Seconds
[021]	sa.sadm_monitor_recent_count	Monitor web page nb. recent scripts	5
[022]	sa.sadm_monitor_recent_exclude	Monitor web page exclude list	sadm_nmon_watcher,sysmon_script_test

----- Email Section -----

[023]	sa.sadm_mail_addr	SADMIN SysAdmin Email(s)	brucetalbot95@gmail.com
[024]	sa.sadm_smtp_server	Your internet smtp server	smtp.gmail.com
[025]	sa.sadm_smtp_port	Your internet smtp server port	587
[026]	sa.sadm_smtp_sender	Your internet smtp email address	brucetalbot95@gmail.com
[027]	sa.sadm_gmpw	Your internet smtp email password	*Hidden*
[028]	sa.sadm_email_startup	Send email to SysAdmin on startup	Y
[029]	sa.sadm_email_shutdown	Send email to SysAdmin on shutdown	Y

----- Monitoring Section -----

[030]	sa.sadm_alert_type	0=NoAlert 1=OnError 3=OnSuccess 4=All	1
[031]	sa.sadm_alert_repeat	0=AlertOnce or Sec. before alert repeat	0 sec
[032]	sa.sadm_alert_group	Error Group Name (Default Group)	default
[033]	sa.sadm_warning_group	Warning Group Name	default
[034]	sa.sadm_info_group	Info Group Name	default
[035]	sa.sadm_ssh_port	Default SSH Port to talk with client	32
[036]	sa.sadm_textbelt_key	TextBelt.com API Key	*Hidden*
[037]	sa.sadm_textbelt_url	TextBelt.com API URL	https://textbelt.com/text
[038]	sa.sadm_ntfy_email	Email related to NTFY	duplessis.jacques@gmail.com
[039]	sa.sadm_ntfy_pwd	Password related to NTFY	5ne46yJFqmQS24rU8e9
[040]	sa.sadm_ntfy_token	Token related to NTFY	tk_2a511lu2o46k5hdg1t64h5u55wt0
[041]	sa.sadm_ntfy_topic	Topic use with NTFY	sadmdev

[042]	sa.sadm_ntfy_url	URL to send notification to NTFY	https://ntfy.sh
[043]	sa.sadm_ntfy_user	NTFY user name	duplessis.jacques@gmail.com

----- Files and Logs pruning Section -----

[044]	sa.sadm_rch_keepdays	Nb. days to keep unmodified .rch file	35 days
[045]	sa.sadm_log_keepdays	Nb. days to keep unmodified .log file	35 days
[046]	sa.sadm_max_rchline	Trim rch file to this max. of lines	30 lines
[047]	sa.sadm_max_logline	Trim log to this maximum of lines	500 lines
[048]	sa.sadm_days_history	Days to keep alert in History file	100 days
[049]	sa.sadm_max_arc_line	Max. lines in alert in History file	1000 lines
[050]	sa.sadm_nmon_keepdays	Nb. of days to keep .nmon perf. file	10 days

----- Subnet Network Scanner Section -----

[051]	sa.sadm_network1	Network/Netmask 1 inv. IP/Name/Mac	192.168.1.0/24
[052]	sa.sadm_network2	Network/Netmask 2 inv. IP/Name/Mac	None
[053]	sa.sadm_network3	Network/Netmask 3 inv. IP/Name/Mac	None
[054]	sa.sadm_network4	Network/Netmask 4 inv. IP/Name/Mac	None
[055]	sa.sadm_network5	Network/Netmask 5 inv. IP/Name/Mac	None

----- ReaR Backup Section -----

[056]	sa.sadm_rear_backup_script	ReaR backup script name	sadm_rear_backup.sh
[057]	sa.sadm_rear_nfs_server	ReaR NFS Server IP or Name	lestrade.maison.ca
[058]	sa.sadm_rear_nfs_server_ver	ReaR NFS Server Version	3
[059]	sa.sadm_rear_nfs_mount_point	ReaR NFS Mount Point	/backup/backup_rear
[060]	sa.sadm_rear_backup_to_keep	ReaR NFS Backup - Nb. to keep	2 copies
[061]	sa.sadm_rear_backup_dif	Alert if pct between cur. & prev. size	25 %
[062]	sa.sadm_rear_backup_interval	Alert if no backup for more than X days	14 days
[063]	sa.sadm_rear_del_failed_backup	Del backup if failed integrity check	Y
[064]	sa.sadm_rear_backup_concurrent	Number of Concurrent rear backup	1
[065]	sa.sadm_rear_batch_mode	Run backup in batch mode (y/N)	N
[066]	sa.sadm_rear_batch_startup_time	Batch startup time	00:10
[067]	sa.sadm_rear_batch_day2run	0=AnyDay, 1=Su,2=Mo,3=Tu,...7=Sa	"2,7"
[068]	sa.sadm_rear_batch_mth2run	0=AnyMth or [1,2,,,12] Month to run	"0"
[069]	sa.sadm_rear_batch_date2run	0=AnyDate, Date to run [1,2...27,28]	"0"

----- Backup Section -----

[070]	sa.sadm_backup_nfs_server	NFS Backup IP or Server Name	batnas.maison.ca
[071]	sa.sadm_backup_nfs_server_ver	NFS Backup Server Version	3
[072]	sa.sadm_backup_nfs_mount_point	NFS Backup Mount Point	/volume1/backup/backup_sadmin
[073]	sa.sadm_backup_dif	Alert if pct between cur. & prev. size	40 %
[074]	sa.sadm_backup_interval	Alert if no backup for more than X days	1 days
[075]	sa.sadm_daily_backup_to_keep	Daily Backup to Keep	2 copies
[076]	sa.sadm_weekly_backup_to_keep	Weekly Backup to Keep	1 copies
[077]	sa.sadm_monthly_backup_to_keep	Monthly Backup to Keep	1 copies
[078]	sa.sadm_yearly_backup_to_keep	Yearly Backup to keep	1 copies
[079]	sa.sadm_weekly_backup_day	Weekly Backup Day (1=Mon,7=Sun)	5
[080]	sa.sadm_monthly_backup_date	Monthly Backup Date (1-28)	1
[081]	sa.sadm_yearly_backup_month	Yearly Backup Month (1-12)	12
[082]	sa.sadm_yearly_backup_date	Yearly Backup Date (1-31)	31
[083]	sa.sadm_backup_concurrent	Number of Concurrent Backup	1
[084]	sa.sadm_backup_batch_mode	Run backup in batch mode (y/N)	N
[085]	sa.sadm_backup_batch_startup_time	Batch startup time	00:05
[086]	sa.sadm_backup_batch_day2run	0=AnyDay, 1=Su,2=Mo,3=Tu,...7=Sa	"2,7"

[087]	sa.sadm_backup_batch_mth2run	0=AnyMth or [1,2,,,12] Month to run	"0"
[088]	sa.sadm_backup_batch_date2run	0=AnyDate, Date to run [1,2...27,28]	"0"

----- Virtual Machines Export Section -----

[089]	sa.sadm_vm_export_script	Virtual Machine Export Script Name	sadm_vm_export.sh
[090]	sa.sadm_vm_export_nfs_server	Virtual Machine Export Script Name	batnas.maison.ca
[091]	sa.sadm_vm_export_nfs_server_ver	NFS Export Server Version	3
[092]	sa.sadm_vm_export_mount_point	NFS Export Mount Point	/volume1/backup/backup_vm/vbox_exports
[093]	sa.sadm_vm_export_dif	Alert if pct between cur. & prev. size	25 %
[094]	sa.sadm_vm_export_to_keep	Nb. of export to keep	2 copies
[095]	sa.sadm_vm_export_interval	Alert if no export for more than X days	21 days
[096]	sa.sadm_vm_export_alert	Issue an alert if interval reached	Y
[097]	sa.sadm_vm_user	User part of 'vboxusers' user group	jacques
[098]	sa.sadm_vm_stop_timeout	Max. seconds given for acpi shutdown	60 seconds
[099]	sa.sadm_vm_start_interval	Sec. to sleep between each VM start	30 seconds
[100]	sa.sadm_vm_export_concurrent	Number of Concurrent Export	1
[101]	sa.sadm_vm_export_batch_mode	Run export in batch mode (y/N)	N
[102]	sa.sadm_vm_export_batch_start_time	Start time for launch batch export	00:01
[103]	sa.sadm_vm_export_batch_day2run	0=AnyDay, 1=Su,2=Mo,3=Tu,...7=Sa	"2,7"
[104]	sa.sadm_vm_export_batch_mth2run	0=AnyMth, Months Numbers 1,2,12	"0"
[105]	sa.sadm_vm_export_batch_date2run	0=AnyDate, Date to run [1,2...27,28]	"0"

----- System Update Section -----

[106]	sa.sadm_osupdate_script	System Update Script Name	sadm_osupdate.sh
[107]	sa.sadm_osupdate_interval	Alert if no update for more than X days	15 days
[108]	sa.sadm_osupdate_autoremove	Auto remove 'orphaned' packages.	N
[109]	sa.sadm_osupdate_flatpak	Update Flatpak packages	N
[110]	sa.sadm_osupdate_snap	Update Snap packages	N
[111]	sa.sadm_osupdate_reboot_needed	Reboot after each update	N
[112]	sa.sadm_osupdate_reboot_time	Sec. given to bring up O/S & app.	240 seconds
[113]	sa.sadm_osupdate_lock	Lock system during the update	Y
[114]	sa.sadm_osupdate_concurrent	Number of Concurrent System Updates	1
[115]	sa.sadm_osupdate_batch_mode	Run O/S Update in Batch Mode (y/N)	N
[116]	sa.sadm_osupdate_batch_start_time	Batch Update Start Time	00:05
[117]	sa.sadm_osupdate_batch_day2run	0=AnyDay, 1=Su,2=Mo,3=Tu,...7=Sa	"2,7"
[118]	sa.sadm_osupdate_batch_mth2run	0=AnyMth, Months Numbers 1,2,12	"0"
[119]	sa.sadm_osupdate_batch_date2run	0=AnyDate, Date to run [1,2...27,28]	"0"

5) Directories variables available to Developers

[001]	sa.dir_base	SADMIN Root Directory	/opt/sadmin
[002]	sa.dir_bin	SADMIN Scripts Directory	/opt/sadmin/bin
[003]	sa.dir_tmp	System Temporary file(s) Directory	/opt/sadmin/tmp
[004]	sa.dir_lib	SADMIN Shell & Python Library Dir.	/opt/sadmin/lib
[005]	sa.dir_log	SADMIN Script Log Directory	/opt/sadmin/log
[006]	sa.dir_cfg	SADMIN Configuration Directory	/opt/sadmin/cfg
[007]	sa.dir_sys	System Startup/Shutdown Script Dir.	/opt/sadmin/sys
[008]	sa.dir_doc	SADMIN Documentation Directory	/opt/sadmin/doc
[009]	sa.dir_pkg	SADMIN Packages Directory	/opt/sadmin/pkg
[010]	sa.dir_dat	System Data Directory	/opt/sadmin/dat
[011]	sa.dir_nmon	System 'nmon'- Data Collected Dir.	/opt/sadmin/dat/nmon
[012]	sa.dir_dr	System Disaster Recovery Info Dir.	/opt/sadmin/dat/dr
[013]	sa.dir_rch	System Return Code History Dir.	/opt/sadmin/dat/rch
[014]	sa.dir_net	System Network Information Dir.	/opt/sadmin/dat/net
[015]	sa.dir_rpt	System Monitor Report Directory	/opt/sadmin/dat/rpt
[016]	sa.dir_dbb	SADMIN Database Backup Directory	/opt/sadmin/dat/dbb
[017]	sa.dir_setup	SADMIN Setup Directory.	/opt/sadmin/setup
[018]	sa.dir_usr	User root directory	/opt/sadmin/usr
[019]	sa.dir_usr_bin	User Scripts Directory	/opt/sadmin/usr/bin
[020]	sa.dir_usr_lib	User Library Directory	/opt/sadmin/usr/lib
[021]	sa.dir_usr_doc	User Documentation Dir.	/opt/sadmin/usr/doc
[022]	sa.dir_usr_mon	User System Monitor Scripts	/opt/sadmin/usr/mon
[023]	sa.dir_www	SADMIN Web Site Root Directory	/opt/sadmin/www
[024]	sa.dir_www_doc	SADMIN Web Site Documentation Dir.	/opt/sadmin/www/doc
[025]	sa.dir_www_dat	SADMIN Web Site Global Data Dir.	/opt/sadmin/www/dat
[026]	sa.dir_www_lib	SADMIN Web Site PHP Library Dir.	/opt/sadmin/www/lib
[027]	sa.dir_www_tmp	SADMIN Web Temp Working Dir.	/opt/sadmin/www/tmp
[028]	sa.dir_www_perf	SADMIN Web Performance Graph Dir.	/opt/sadmin/www/tmp/perf

6) SADMIN File names use and set by the Python library, for you to use

----- Available File Name Aliases Variables -----

[001]	sa.pid_file	Current script PID file	/opt/sadmin/tmp/sadmlib_std_demo.pid
[002]	sa.cfg_file	SADMIN Main Configuration file	/opt/sadmin/cfg/sadmin.cfg
[003]	sa.cfg_hidden	SADMIN configuration template file	/opt/sadmin/cfg/.sadmin.cfg
[004]	sa.alert_file	Alert Group definition file	/opt/sadmin/cfg/alert_group.cfg
[005]	sa.alert_init	Alert Group definition template file	/opt/sadmin/cfg/.alert_group.cfg
[006]	sa.alert_hist	Alert History file	/opt/sadmin/cfg/alert_history.txt
[007]	sa.alert_hini	Alert History template file	/opt/sadmin/cfg/.alert_history.txt
[008]	sa.tmp_file1	User usable Temp Work File 1	/opt/sadmin/tmp/tmp_file_1.368820
[009]	sa.tmp_file2	User usable Temp Work File 2	/opt/sadmin/tmp/tmp_file_2.368820
[010]	sa.tmp_file3	User usable Temp Work File 3	/opt/sadmin/tmp/tmp_file_3.368820
[011]	sa.log_file	Log File Name	/opt/sadmin/log/bilbo_sadmlib_std_demo.log
[012]	sa.rch_file	Return Code History File Name	/opt/sadmin/dat/rch/bilbo_sadmlib_std_demo.rch
[013]	sa.dbpass_file	SADMIN Database User Password File	/opt/sadmin/cfg/.dbpass
[014]	sa.rpt_file	[SYS]tem [MON]itor report file	/opt/sadmin/dat/rpt/bilbo.rpt
[015]	sa.backup_list	Backup file list	/opt/sadmin/cfg/backup_list.txt
[016]	sa.backup_list_init	Backup file template file	/opt/sadmin/cfg/.backup_list.txt
[017]	sa.backup_exclude	Backup exclude list file name	/opt/sadmin/cfg/backup_exclude.txt
[018]	sa.backup_exclude_init	Backup exclude template file	/opt/sadmin/cfg/.backup_exclude.txt

7) Command Path set and use by SADMIN Python Library for more security.

```
-----
[001] sa.cmd_dmidcode          Cmd. 'dmidecode', Get model & type      /usr/sbin/dmidecode
[002] sa.cmd_bc                Cmd. 'bc', Do some Math.                  /usr/bin/bc
[003] sa.cmd_fdisk             Cmd. 'fdisk', Get Partition Info          /usr/sbin/fdisk
[004] sa.cmd_which             Cmd. 'which', Get Command location        /usr/bin/which
[005] sa.cmd_perl              Cmd. 'perl', epoch time Calc.            /usr/bin/perl
[006] sa.cmd_mutt              Cmd. 'mutt', Used to Send Email           /usr/bin/mutt
[007] sa.cmd_curl              Cmd. 'curl', To send alert to Slack       /usr/bin/curl
[008] sa.cmd_lscpu             Cmd. 'lscpu', Socket & thread info        /usr/bin/lscpu
[009] sa.cmd_nmon              Cmd. 'nmon', Collect Perf Statistic      /usr/bin/nmon
[010] sa.cmd_parted            Cmd. 'parted', Get Disk Real Size        /usr/sbin/parted
[011] sa.cmd_ethhtool          Cmd. 'ethhtool', Get System IP Info      /usr/sbin/ethhtool
[012] sa.cmd_rrdtool           Cmd. 'rrdtool' to produce graph          /usr/bin/rrdtool
[013] sa.cmd_lsb_release       Cmd. 'lsb_release' to get dist. info     /usr/bin/lsb_release
[014] sa.cmd_inxi              Cmd. 'inxi' binary location             /usr/bin/inxi
[015] sa.cmd_ssh               Cmd. 'ssh', SSH to SADMIN client         /usr/bin/ssh
-----
```

```
=====
Script exit code is 0 (Success) and execution time is 00:00:01
History file $SADMIN/dat/rch/bilbo_sadmlib_std_demo.rch trim to 50 lines.
Send alert only when it terminate with error.
Log file $SADMIN/log/bilbo_sadmlib_std_demo.log trim to 500 lines.
Fri 17 Jul 2026 16:34:19 - End of sadmlib_std_demo.py
=====
```